Metal Gear Solid 4: Guns of the Patriots
MGS 4 might not look very impressive in screenshots but you have to watch this baby in motion to truly appreciate the superior animations, the filters used and the smoke and fire effects

Killzone 2
Killzone uses a Deferred Rendering engine that integrates multisample anti-aliasing and uses the PlayStation 3's cell process to process geometry and indirect lighting in parallel. Result? Yummy!

Resistance 2
Resistance 2 is all kinds of awesome: not only does it look gorgeous but it also features a co-operative campaign which supports up to 8 players online or 2 players split-screen, with classes including a medic class, a heavy weapons class, and a special ops class. Its online multiplayer will support up to 60 players!

anisotropic filtering and running at higher resolutions. Let's see *Crysis* running on a console.

Reality: Piracy Is Rampant On The PC, Retail Sales Are Dwindling For The PC Market

This is why we see the game developers flocking to the consoles, the reason why quality PC releases are few and far between. So, if there is anyone to blame for the sorry state of PC releases, it's us. If you like games, buy them.

Another angle to consider here, is that of online distribution. Developers on the PC are increasingly taking to an online model—such as Steam—to distribute their content. This has benefits of cheaper software, a better control over piracy, and the ability to distribute to far reaches of the world, without the need for a physical distribution chain. Online distribution also accounts for some drop in sales. Nevertheless, the fact remains that piracy on the PC is hurting the PC game developer, and he's looking to the consoles to see better returns. This exodus could mean more delayed releases on the PC, while the consoles get a game first, or worse, developers dropping off the PC bandwagon altogether. The latter though, is unlikely to happen, or so we hope.

Reality: It's About The Games

PC or console, this generation is all about the games, and what a glorious generation this promises to be! We have already seen some great titles on the consoles—*Gears of War*, *Halo 3*, *Uncharted*, *Ratchet and Clank: Tools of Destruction*, *Gran Turismo 5: Prologue*, *Grand Theft Auto 4*, *Crysis*, *Age of Conan: Hyborian Adventures*, *Mario Galaxy*, with many more to come—*LittleBigPlanet*, *Alan Wake*, *FarCry 2*, *Resistance 2*, *Killzone 2*, *Fable 2*, *StarCraft 2*, *Duke Nukem Forever* (we hope!), *Diablo III* (oh please, Mr. Blizzard!) and many many more. There has never been a better time to be a gamer. So here's our suggestion for the year ahead—enjoy!

The Road To Realism

Between *Gears of War*, *Crysis* and the upcoming *Metal Gear Solid 4*, this generation of games have been a visual feast. Recall the screenshots comparing in-game scenes of *Crysis* to real-world locales—could you really tell the difference? Did your jaw not drop when you saw the cars in *Gran Turismo 5*? We are inching ever so close to the holy grail of computer graphics resembling the real world. There are still some huge hurdles to clear. The biggest is psychological—the so called Uncanny Valley hypothesis, that "when computer-rendered humans look and act almost, but not entirely, like actual humans, it causes a response of revulsion among human observers." As we get closer to human-like, and life-like, we being to notice the tiny flaws—the unnatural gait, the lip movement, the eye flutter, the way the bush reacts to our passing, our in-game shadows, and so on. Another hurdle is that of texture quality—a game looks as good as the textures it can display. High-quality textures take up both disc space and graphics card bandwidth and memory—all are in limited supply, finite, even though Blu-Ray discs slightly alleviate one of the problems. One possible solution is the *MegaTexture* approach taken by id Software, recently demonstrated by John Carmack as part of id's upcoming Tech 5 game engine: the scene demonstrated had 20 GB of texture data, using textures with up to 128000 x 128000 pixel resolution, and a completely dynamically changeable world.

MegaTexture employs an extremely large terrain texture, instead of repeating multiple smaller textures, which adds to the visual fidelity of a scene and makes the artist's work much easier to boot. The single large texture is stored on removable media and streamed as needed, allowing large amounts of detail and variation over a large area.

The tech also alleviates the uncanny valley associated with shadows as the Tech 5 renderer includes a penumbra in the shadowing, by using shadow maps. The engine will also support multiple processors by multi-threading many of its tasks such as rendering, some gameplay logic, AI, physics and sound processing.

Multi-processing is also been appropriately utilised by the upcoming *Alan Wake*. This psychological action thriller will showcase some great effects due to its leveraging of multiple processors—including a complete modelling of atmospheric scattering—giving us realistic cloud formations and skies, day / night time cycles, ambient occlusion, normal mapping, high dynamic rendering, fully volumetric shadows that are projected through the entire world, full weather modelling, bloom, depth of field and a draw distance of around 2 kilometres! The game is being designed with five threads—the rendering thread will prepare vertices to be sent to the GPU, an audio thread and a streaming thread, which will be used to stream data off of the DVD or hard disk as well as decompress the data on the fly and allowing seamless traversal of environments without loading screens, a thread for physics calculations, which are being simulated using the Havoc engine, and one for terrain tessellation. Moreover, an entire core processor can be dedicated to physical calculations alone.

Games are only going to improve in visual quality as we progress, the next few years should bring us eye candy unlike any we could imagine. If you took a look at what's around you now, you'd get an idea of the awesomeness that is to follow... ■

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Crysis
With over one million copies sold worldwide, *Crysis* for the PC redefined what a game could offer as visuals

Far Cry 2
Far Cry 2 will take place in a sprawling African region—covering everything from small towns to the savannah. The game features destructible environments and foliage, vehicles, and a dynamic weather system which changes from sunny skies to storms depending on how well the player is doing in the game!

LittleBigPlanet
LittleBigPlanet proves that you don't need guns to look good. This PS3 game innovates with both its look and its gameplay: players shape and develop the game on-the-fly to build custom levels either individually, collaboratively, or competitively. The game's core focus in on co-operative, and physics-based platforming

Gran Turismo 5
Gran Turismo 5: Prologue for the PlayStation 3 brings the most realistically detailed cars to a racing game, till date.